Inequality Is No Sufficient Statistic

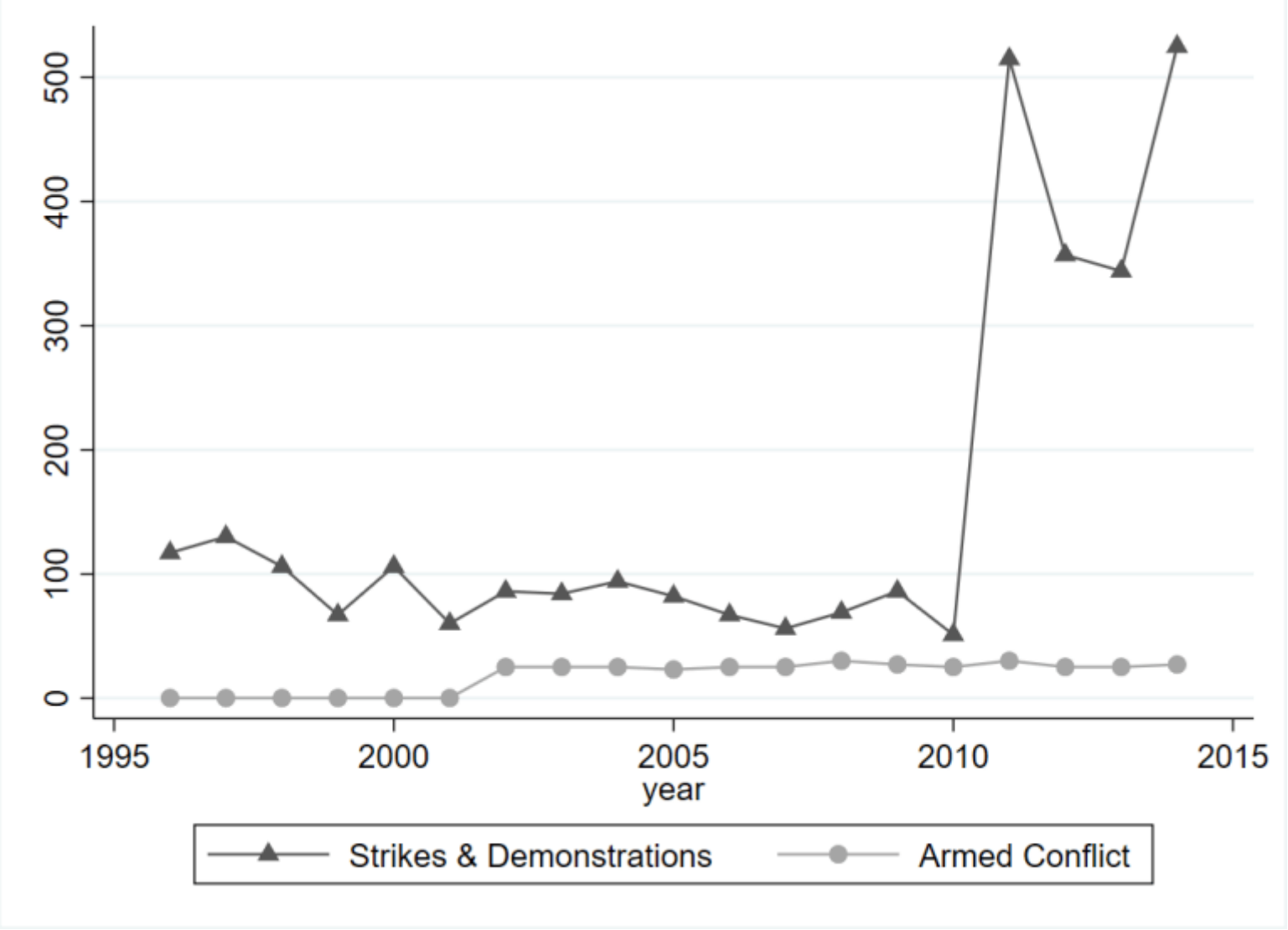
IEA RIDGE Lecture

by Danny Quah

Sat 03 Jul 2021

NUS National University of Singapore

Lee Kuan Yew School of Public Policy



Source: C. Witte et al (2020).

“Inequality Is No Sufficient Statistic”

Our greatest fears and concerns

Resource-Inadequacy vs Disparity

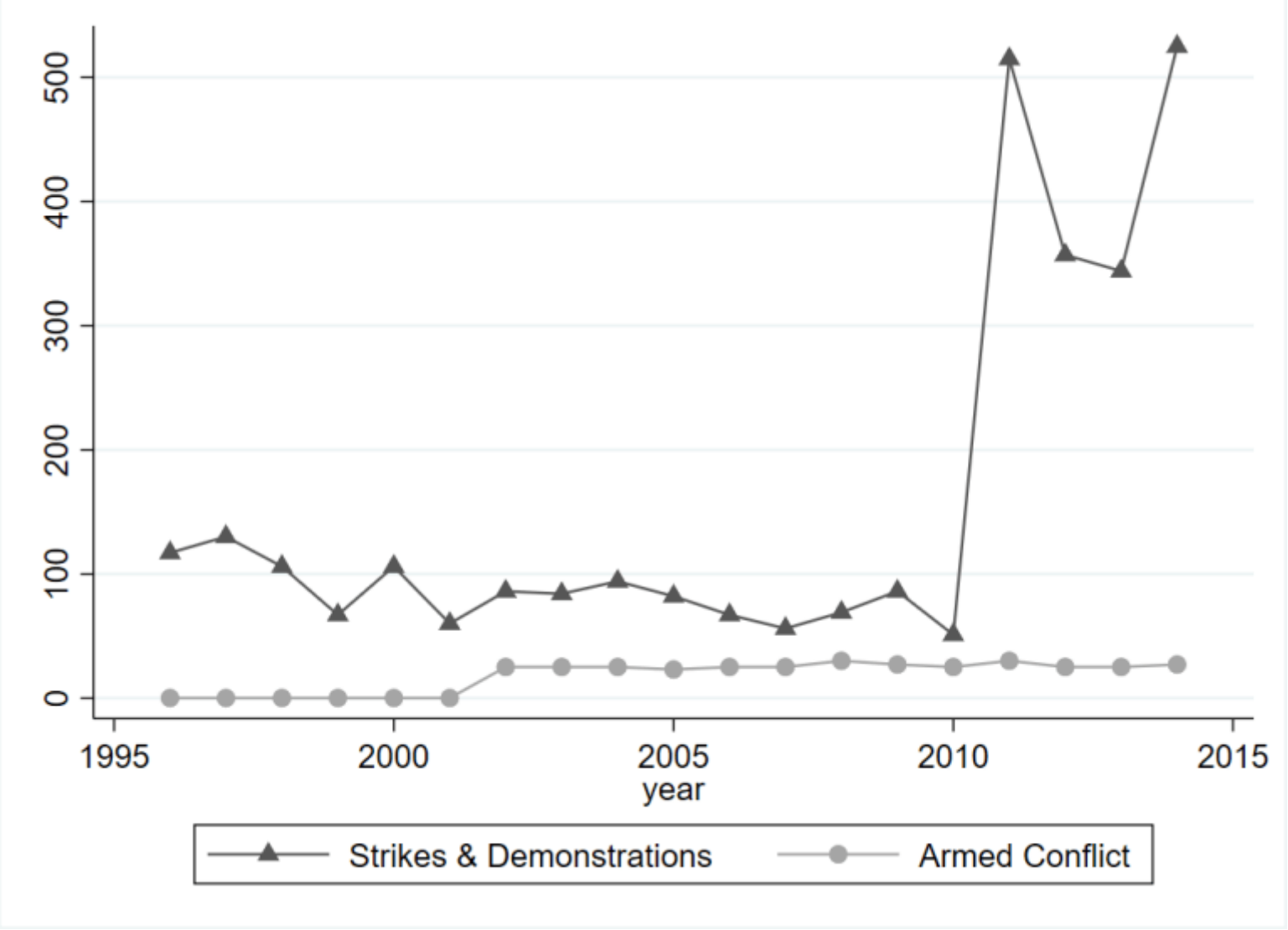
Empirical Evidence

“Inequality Is No Sufficient Statistic”

Our greatest fears and concerns

Resource-Inadequacy vs Disparity

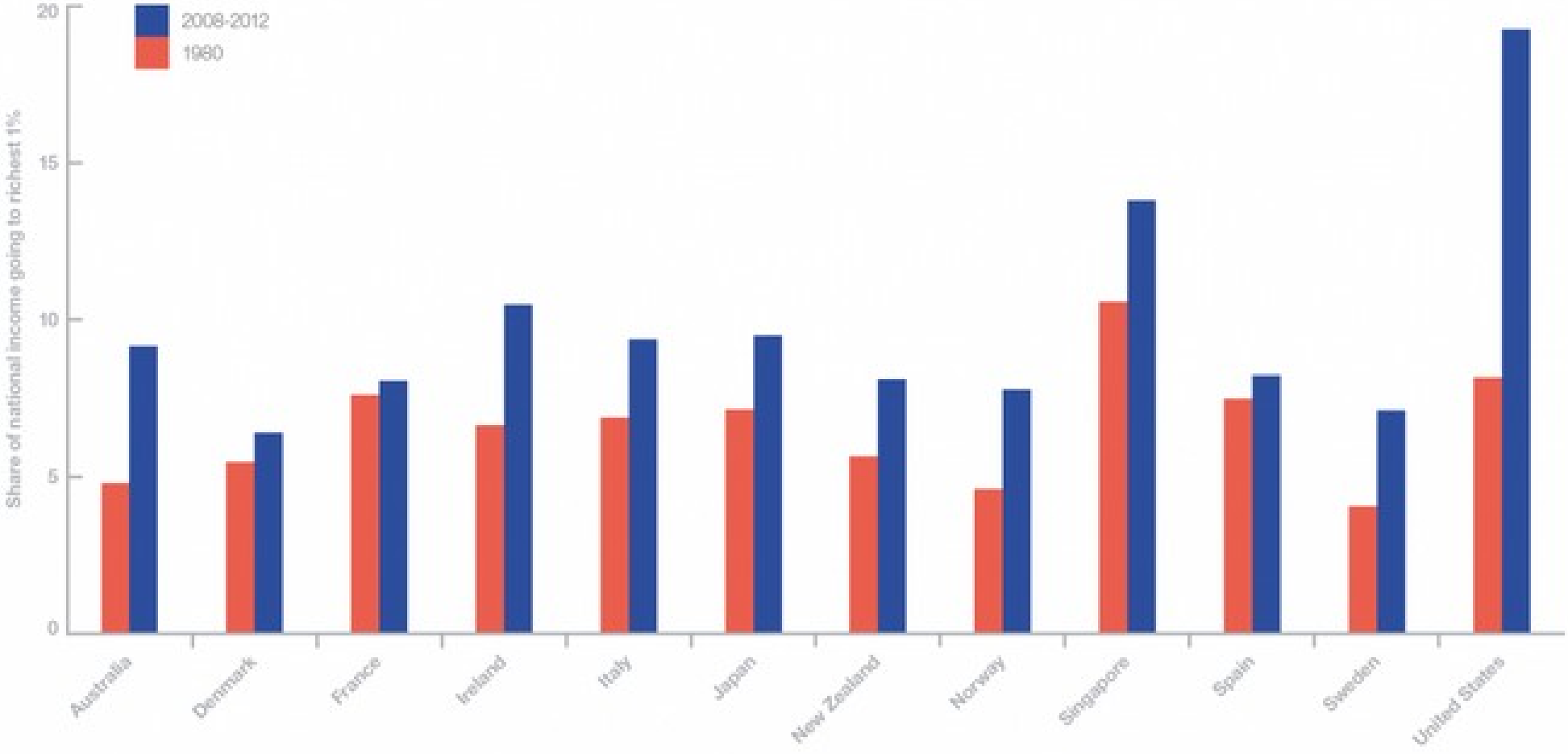
Empirical Evidence



Source: C. Witte et al (2020).

US Inequality

Source: Saez and Piketty



Source: F. Alvaredo, A. B. Atkinson, T. Piketty, and E. Saez, 2013. 'The World Top Incomes Database'

Hypotheses

1. "Inequality immiserises the disadvantaged in society and, through obstructing social mobility, frustrates expectations of a better life. All these cause social discord."

2. "Inequality-driven social discord produces political upheaval."

Emerging Grand Narrative

"Globalisation and technological change have polarised society into a small elite with highly paid, secure jobs on one side, and on the other side are growing numbers of people, including an increasingly squeezed middle class, in insecure, poorly-paid work."

This growing inequality has resulted in "erosion of solidarity, social trust and faith in democratic institutions".

(????)

Unanimity

"Inequality is the root of social evil"

(Pope Francis, 2014)

" 不患寡而患不均 " "Disturbing is not scarcity but inequality"

(Confucius)

The greatest obstacle in "the defining challenge of our time"

(Barack Obama, 2013)

The first step to combat inequality is to understand it

Promoting the study of economics enables citizens to understand why inequities are generated and how equity and social mobility can be promoted. *Interview.* ***Español*** ***Português***

Manuel Serrano

João Pedro Santos

4 February 2020

Recife, the Brazilian capital of social inequality. | Wikimedia Commons. Public Domain

Horacio Villalobos/Corbis via Getty Images

Tackling Inequality Is a Political Choice

Jul 15, 2019 | **MAHMOUD MOHIELDIN CAROLINA SÁNCHEZ-PÁRAMO**

Global inequality reveals policy failings

Political will is inexcusably weak, and mere statistics are failing to prod governments beyond election-year soundbites and platitudes.

WED, JUN 06, 2018 - 5:50 AM

India ranks bottom of Oxfam world inequality index

LONDON: India has been ranked among the bottom 11 countries in a new worldwide index on the commitment of different nations to reduce inequalities in their populations.

Posted: Oct 10, 2018 05:28 PM **Updated:** 1 year ago

India has been racked 147th among 157 countries analysed.

n agriculture is declining without a commensurate rise in urban manufacturing
P

BREAKING

04:31 PM Hong Kong: Stocks end the day lower

04:23 PM Europe: Stocks drop at open

04:19 PM Death of Chinese doctor fuels anger, demands for change

04:15 PM Nippon Steel may halt Wakayama blast furnace as competition looms: report

04:13 PM Australia's iron ore ports

U.S. income inequality at highest level in 50 years, economic gap growing in heartland

Six states with the biggest yearly gains in inequality were primarily in the heartland – Alabama, Arkansas, Kansas, Nebraska, New Hampshire and New Mexico.

Homeless sleeping on the streets on Sept. 20, 2019, in downtown Los Angeles, California, the state with the largest homeless population in America. Frederic J. Brown / AFP - Getty Images

Left to own devices?

Kids from different families have uneven access to space, tech support and help with schoolwork. **Venessa Lee** and **Stephanie Yeo** examine how home-based learning shows up the digital divide between the haves and have-nots. **C4&5**

(Left) Housewife Salmiah Salam, 37, and her four kids, aged nine to 14; (Right) Dentist Justin Stewart Saga, 43, and sleep trainer Zoe Chu, 40, with their children, aged three to 13.

Straits Times, Apr 18 2020

DESIGN: SALLY LAM PHOTOS: ALPHONSUS CHERN, MARK CHEONG, COURTESY OF ANGELINA TAN, LIAN

“Inequality Is No Sufficient Statistic”

Our greatest fears and concerns

Resource-Inadequacy vs Disparity

Empirical Evidence

Resource-Inadequacy vs Disparity

"In every country many people have little prospect for a better future. Lacking hope, purpose, or dignity, they watch from society's sidelines as they see others pull ahead to ever greater prosperity. Worldwide many have escaped extreme poverty but even more have neither the opportunities nor the resources to control their lives."

(HDR 2019)

But what will repair this? Reducing inequality? Or allowing poor people resources?

Left to own devices?

Kids from different families have uneven access to space, tech support and help with schoolwork. **Venessa Lee** and **Stephanie Yeo** examine how home-based learning shows up the digital divide between the haves and have-nots. **C4&5**

(Left) Housewife Salmiah Salam, 37, and her four kids, aged nine to 14; (Right) Dentist Justin Stewart Saga, 43, and sleep trainer Zoe Chu, 40, with their children, aged three to 13.

Straits Times, Apr 18 2020

DESIGN: SALLY LAM PHOTOS: ALPHONSUS CHERN, MARK CHEONG, COURTESY OF ANGELINA TAN, LIAN

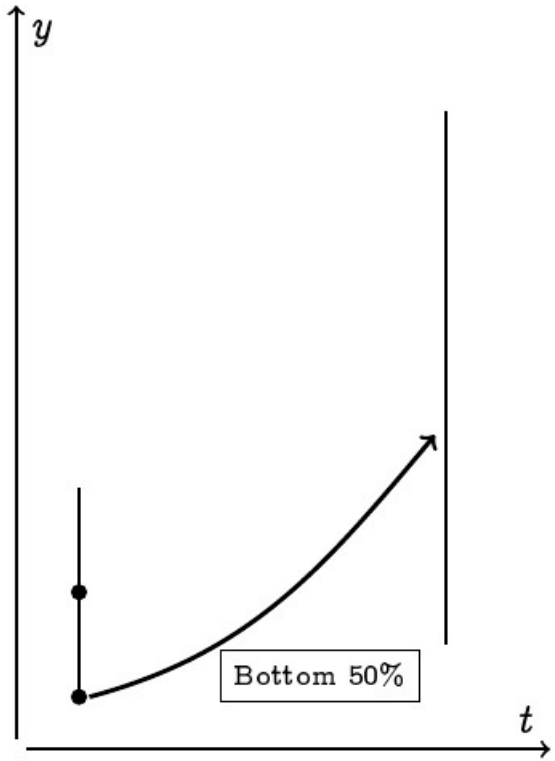
Straits Times, Apr 18 2020

“Inequality Is No Sufficient Statistic”

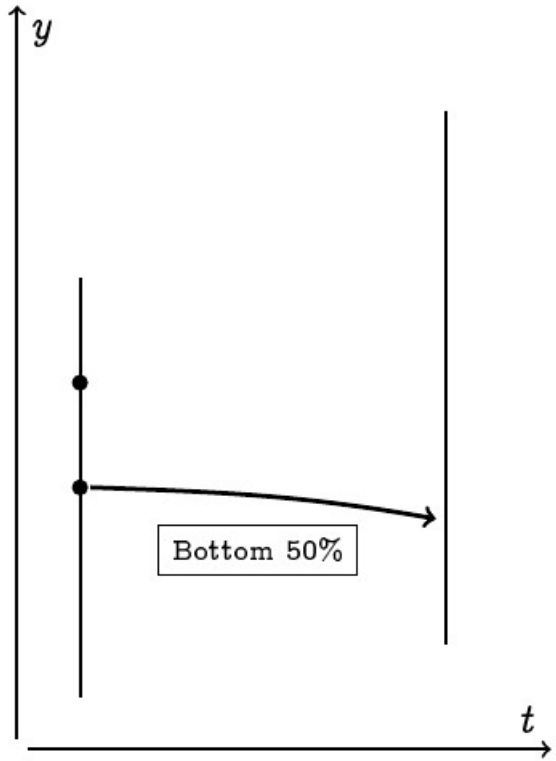
Our greatest fears and concerns

Resource-Inadequacy vs Disparity

Empirical Evidence



(a)



(b)

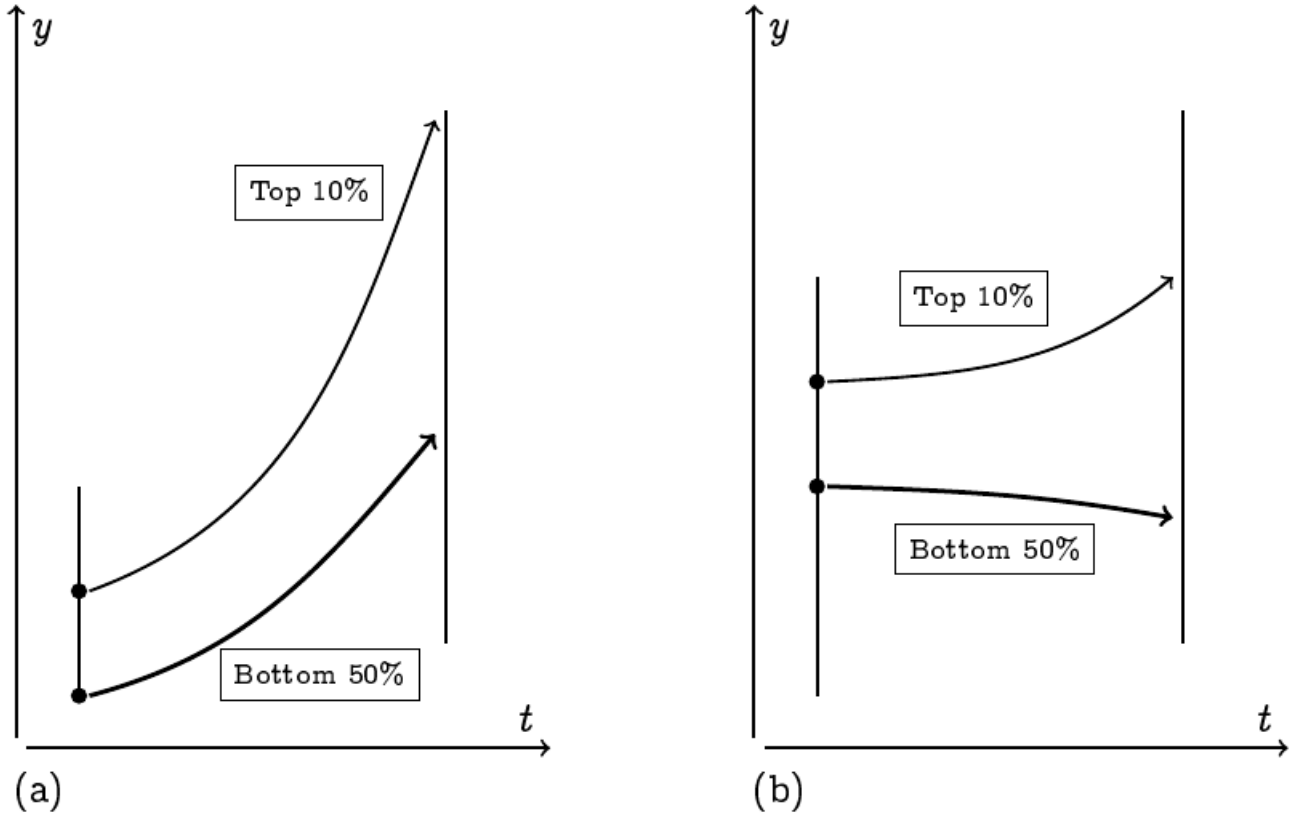
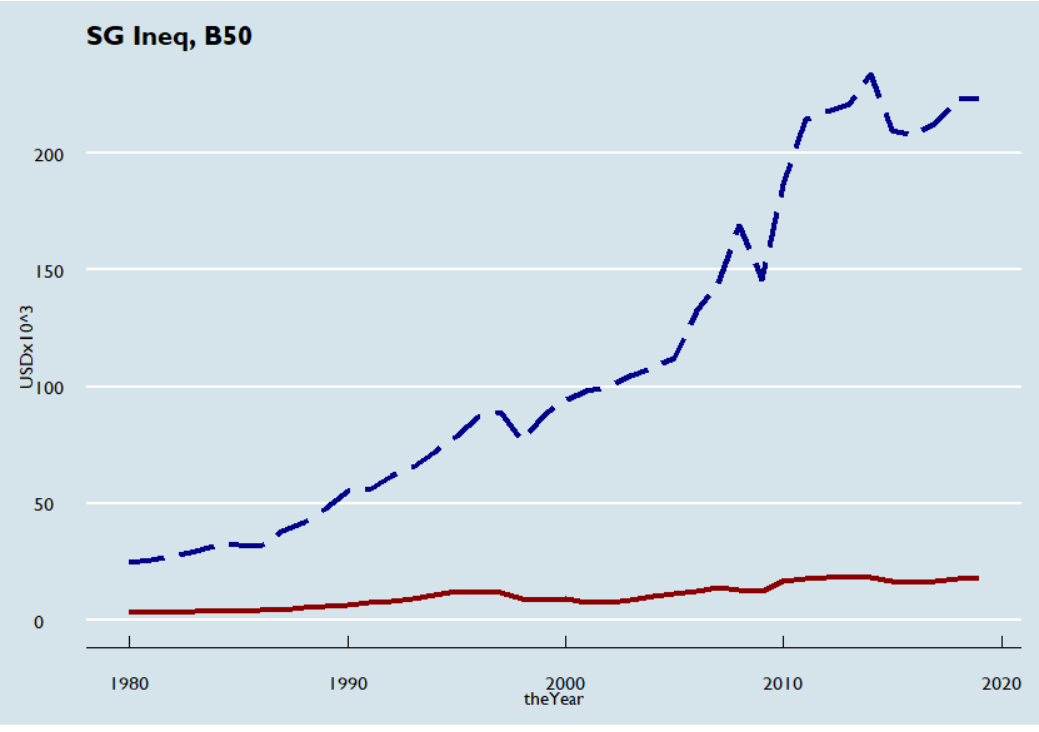
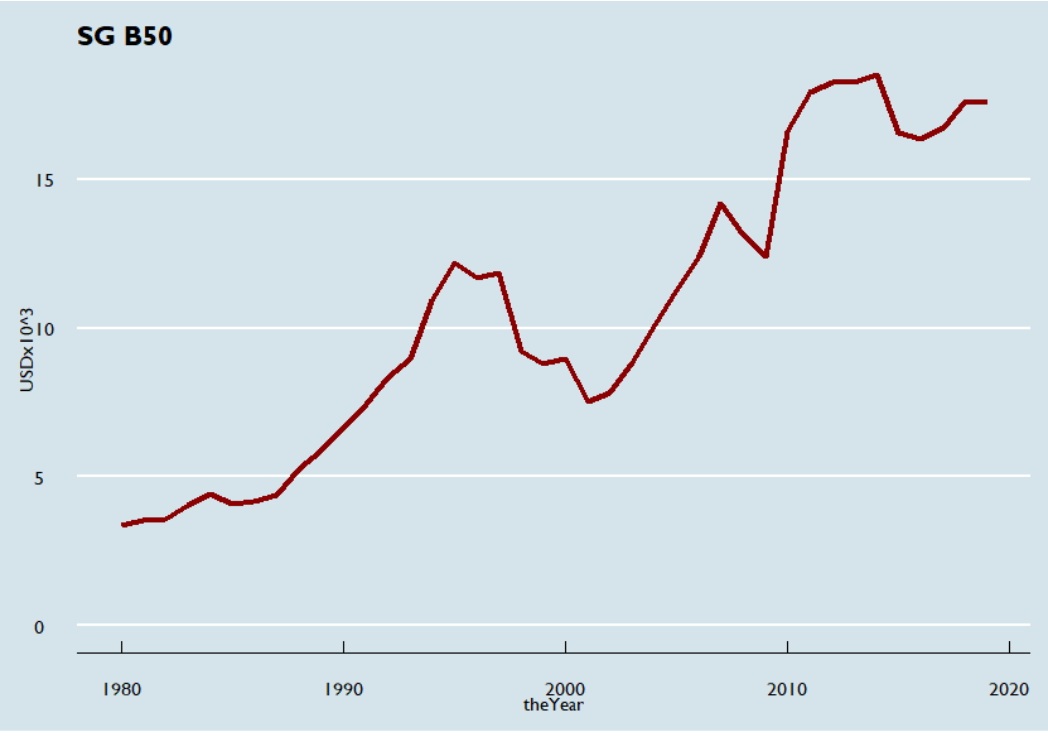


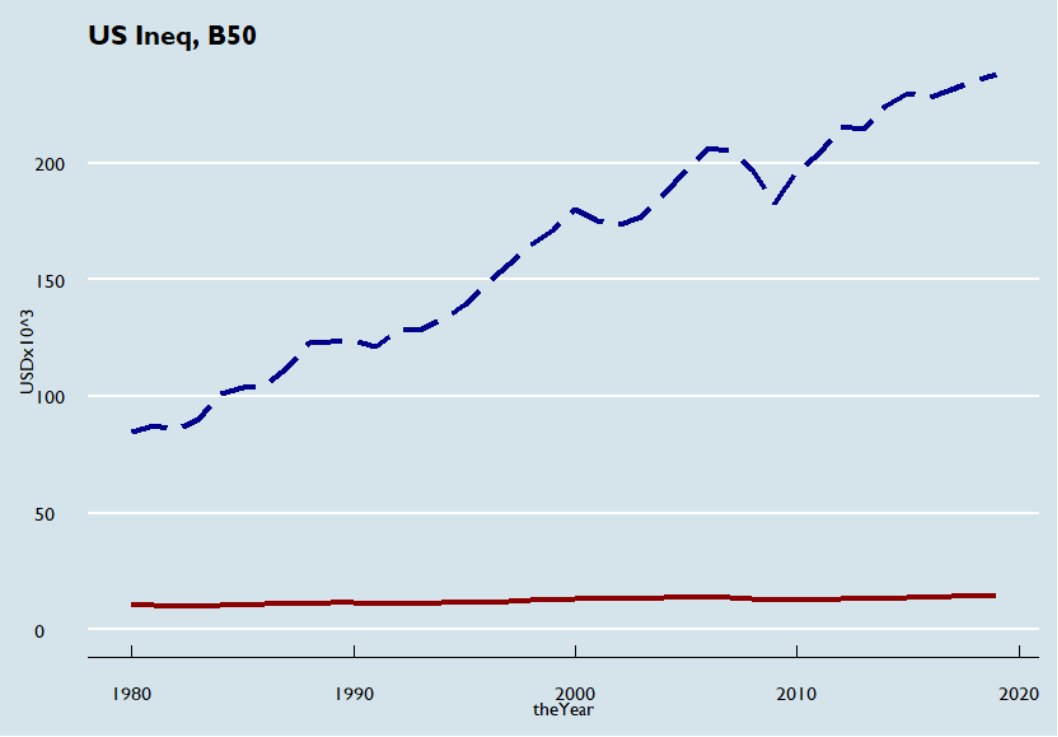
FIGURE 1. GROWTH, INEQUALITY, AND THE LEFT-BEHIND. WHAT DO INEQUALITY AND RICH-INCOME DYNAMICS IMPLY OF OUTCOMES FOR THE POOR?

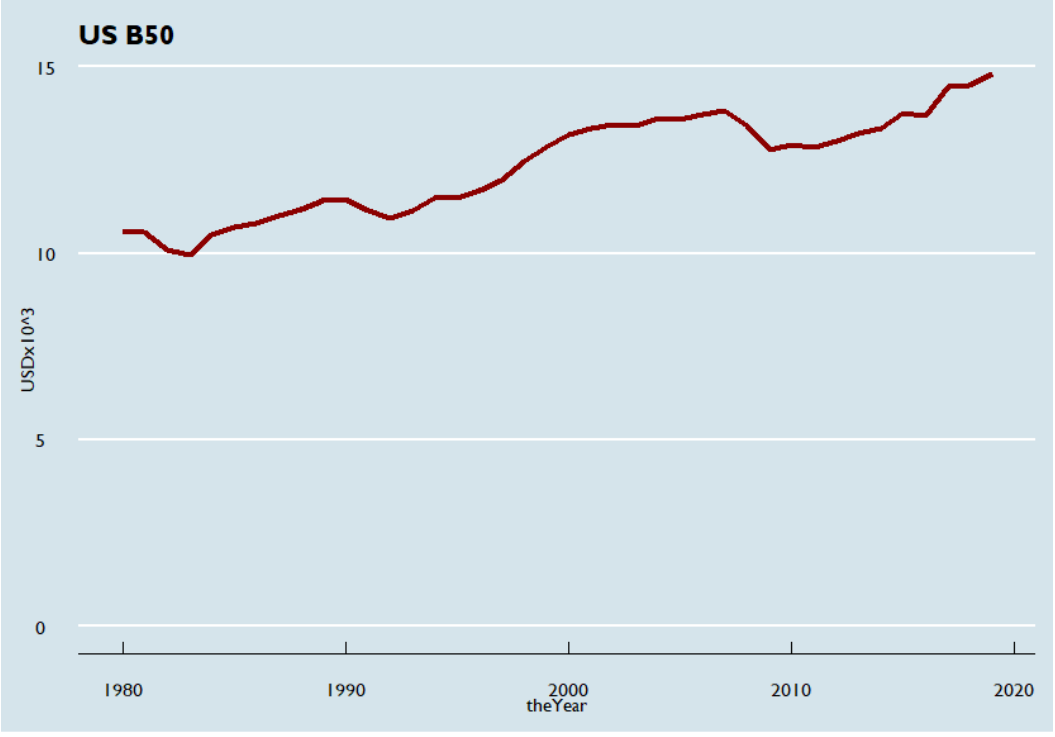
Singapore


SG Ineq, B50
200
150
100
50
0
USDx10^3
1980
1990
2000
2010
2020
theYear

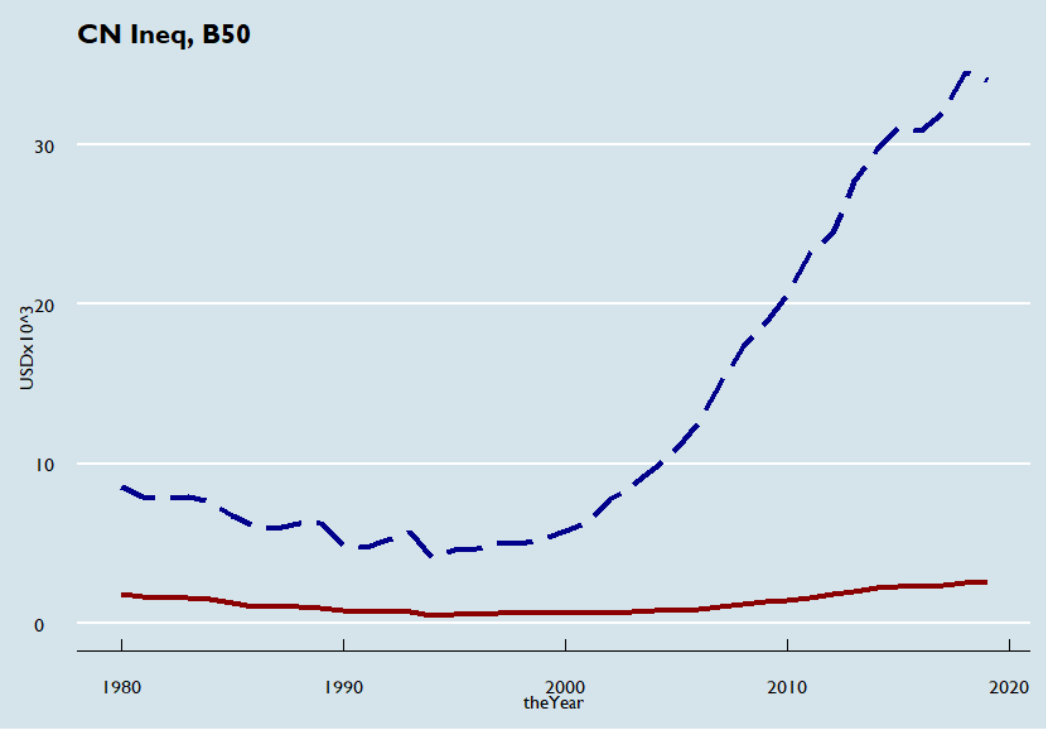

SG B50
15
10
5
0
USDx10^3
1980
1990
2000
2010
2020
theYear

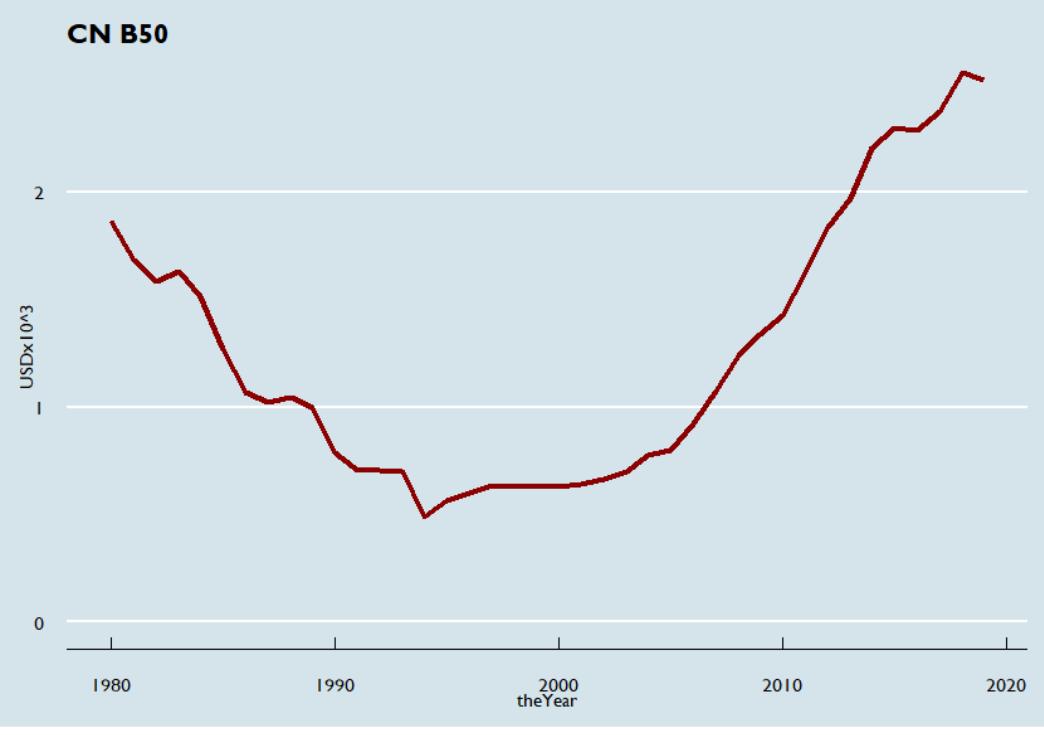
US


US Ineq, B50
200
150
100
50
0
USDx10^3
1980
1990
2000
2010
2020
theYear


US B50
15
10
5
0
USDx10^3
1980
1990
2000
2010
2020
theYear

China


CN Ineq, B50
30
20
10
0
USDx10^3
1980
1990
2000
2010
2020
theYear

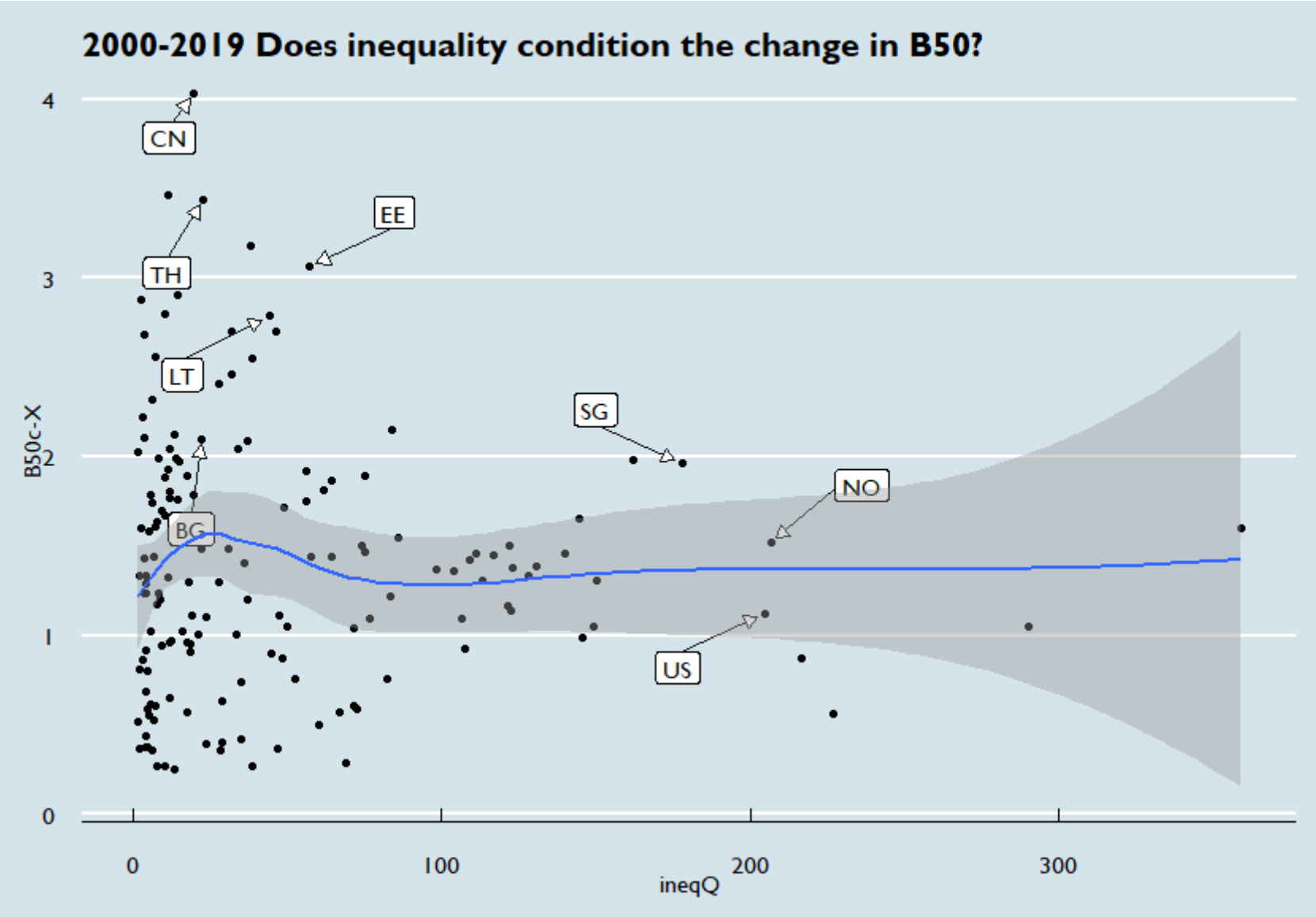

CN B50
2
1
0
USDx10^3
1980
1990
2000
2010
2020
theYear

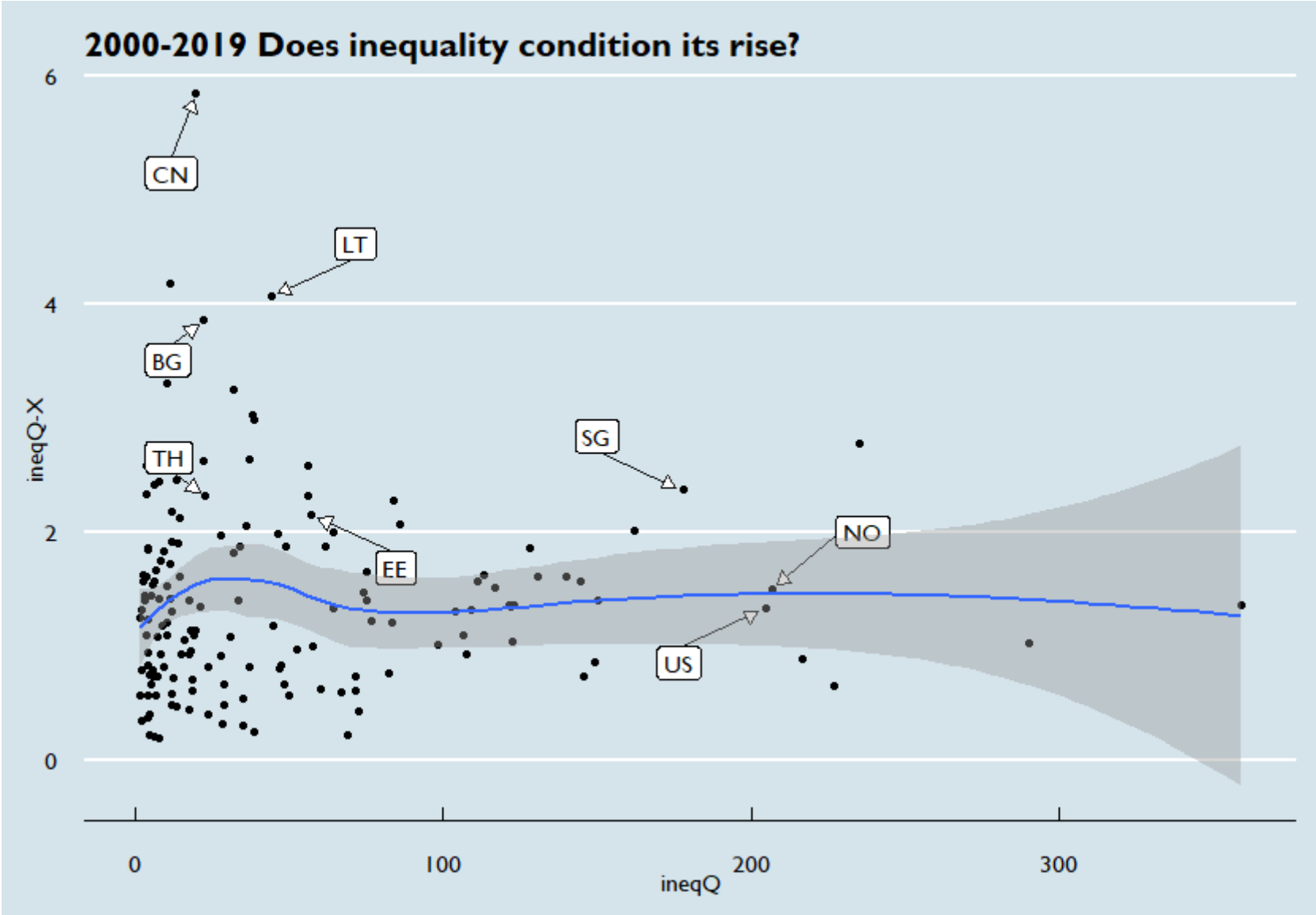
Inequality and Income Mobility

	SG	US	CN
1980-2000			
B50	2.7x (4.9% pa)	1.2x (1.0% pa)	0.3x (-5.5% pa)
Inequality	3.8x	2.1x	0.7x
2000-2019			
B50	2.0x (3.5% pa)	1.1x (0.6% pa)	4.0x (7.3% pa)
Inequality	2.4x	1.3x	5.7x

1. Inequality changes and income mobility go hand in hand
2. Inequality multiples are amplified compared to income mobility

2000-2019 When the bottom 50% rise, so too inequality
ineqQ-X
B50c-X
0
2
4
6
1
2
3
4
CN
BG
LT
SG
TH
NO
EE
US


2000-2019 Does inequality condition the change in B50?
B50c-X
ineqQ
0
1
2
3
4
0
100
200
300
CN
TH
EE
LT
BG
SG
NO
US


2000-2019 Does inequality condition its rise?
6
4
2
0
ineqQ-X
CN
LT
BG
TH
SG
EE
NO
US
0
100
200
300
ineqQ

Correlation

		Correlation	95% Confidence Interval	t(154)
Income Mobility	**Inequality Rise**	0.84	(0.79, 0.88)	19.45
Income Mobility	**Inequality Level**	-0.03	(-0.19, 0.13)	-0.40
Inequality Rise	**Inequality Level**	0.01	(-0.15, 0.17)	0.14

1. Of 99 economies where inequality rose, B50 increased in 95 of them;
2. Of 108 economies where B50 increased, inequality also rose in 88% of them.

The first step to combat inequality is to understand it

Promoting the study of economics enables citizens to understand why inequities are generated and how equity and social mobility can be promoted. *Interview.* ***Español*** ***Português***

Manuel Serrano

João Pedro Santos

4 February 2020

Recife, the Brazilian capital of social inequality. | Wikimedia Commons. Public Domain

Horacio Villalobos/Corbis via Getty Images

Tackling Inequality Is a Political Choice

Jul 15, 2019 | **MAHMOUD MOHIELDIN CAROLINA SÁNCHEZ-PÁRAMO**

The first step to combat inequality is to understand it

Promoting the study of economics enables citizens to understand why inequities are generated and how equity and social mobility can be promoted. *Interview.* *Español Português*

Manuel Serrano

João Pedro Santos

4 February 2020

Fail

Recife, the Brazilian capital of social inequality. | Wikimedia Commons. Public Domain

Horacio Villalobos/Corbis via Getty Images

Tackling Inequality Is a Political Choice

Jul 15, 2019 | MAHMOUD MOHIELDIN CAROLINA SÁNCHEZ-PÁRAMO

Global inequality reveals policy failings

Political will is inexcusably weak, and mere statistics are failing to prod governments beyond election-year soundbites and platitudes.

WED, JUN 06, 2018 - 5:50 AM

India ranks bottom of Oxfam world inequality index

LONDON: India has been ranked among the bottom 11 countries in a new worldwide index on the commitment of different nations to reduce inequalities in their populations.

Posted: Oct 10, 2018 05:28 PM **Updated:** 1 year ago

India has been racked 147th among 157 countries analysed.

n agriculture is declining without a commensurate rise in urban manufacturing
P

BREAKING

04:31 PM Hong Kong: Stocks end the day lower

04:23 PM Europe: Stocks drop at open

04:19 PM Death of Chinese doctor fuels anger, demands for change

04:15 PM Nippon Steel may halt Wakayama blast furnace as competition looms: report

04:13 PM Australia's iron ore ports

U.S. income inequality at highest level in 50 years, economic gap growing in heartland

Six states with the biggest yearly gains in inequality were primarily in the heartland – Alabama, Arkansas, Kansas, Nebraska, New Hampshire and New Mexico.

Homeless sleeping on the streets on Sept. 20, 2019, in downtown Los Angeles, California, the state with the largest homeless population in America. Frederic J. Brown / AFP - Getty Images

Global inequality reveals policy failings

Political will is inexcusably weak, and mere statistics are failing to prod governments beyond election-year soundbites and platitudes.

WED, JUN 06, 2018 - 5:50 AM

India ranks bottom of Oxfam world inequality index

LONDON: India has been ranked among the bottom 11 countries in a new worldwide index on the commitment of different nations to reduce inequalities in their populations.

Posted: Oct 10, 2018 05:28 PM **Updated:** 1 year ago

India has been racked 147th among 157 countries analysed.

Fail

BREAKING

04:31 PM Hong Kong: Stocks end the day lower

04:23 PM Europe: Stocks drop at open

04:19 PM Death of Chinese doctor fuels anger, demands for change

04:15 PM Nippon Steel may halt Wakayama blast furnace as competition looms: report

04:13 PM Australia's iron ore ports

U.S. income inequality at highest level in 50 years, economic gap growing in heartland

Six states with the biggest yearly gains in inequality were primarily in the heartland – Alabama, Arkansas, Kansas, Nebraska, New Hampshire and New Mexico.

Homeless sleeping on the streets on Sept. 20, 2019, in downtown Los Angeles, California, the state with the largest homeless population in America. Frederic J. Brown / AFP - Getty Images

Left to own devices?

Kids from different families have uneven access to space, tech support and help with schoolwork. **Venessa Lee** and **Stephanie Yeo** examine how home-based learning shows up the digital divide between the haves and have-nots. **C4&5**

(Left) Housewife Salmiah Salam, 37, and her four kids, aged nine to 14; (Right) Dentist Justin Stewart Saga, 43, and sleep trainer Zoe Chu, 40, with their children, aged three to 13.

Straits Times, Apr 18 2020

DESIGN: SALLY LAM PHOTOS: ALPHONSUS CHERN, MARK CHEONG, COURTESY OF ANGELINA TAN, LIAN

Fail

Left to own devices?

Kids from different families have uneven access to space, tech support and help with schoolwork. Venessa Lee and Stephanie Yeo examine how home-based learning shows up the digital divide between the haves and have-nots. C4&5

(Left) Housewife Salmiah Salam, 37, and her four kids, aged nine to 14; (Right) Dentist Justin Stewart Saga, 43, and sleep trainer Zoe Chu, 40, with their children, aged three to 13.

Straits Times, Apr 18 2020

DESIGN: SALLY LAM PHOTOS: ALPHONSUS CHERN, MARK CHEONG, COURTESY OF ANGELINA TAN, LIAN

Resource-Inadequacy vs Disparity

Starvation, child hunger, and inter-generational recidivism are problems of poverty and immobility, not inequality: Individual characteristics, not group traits

What makes for enabled, meaningful, and satisfying lives?

Conclusion

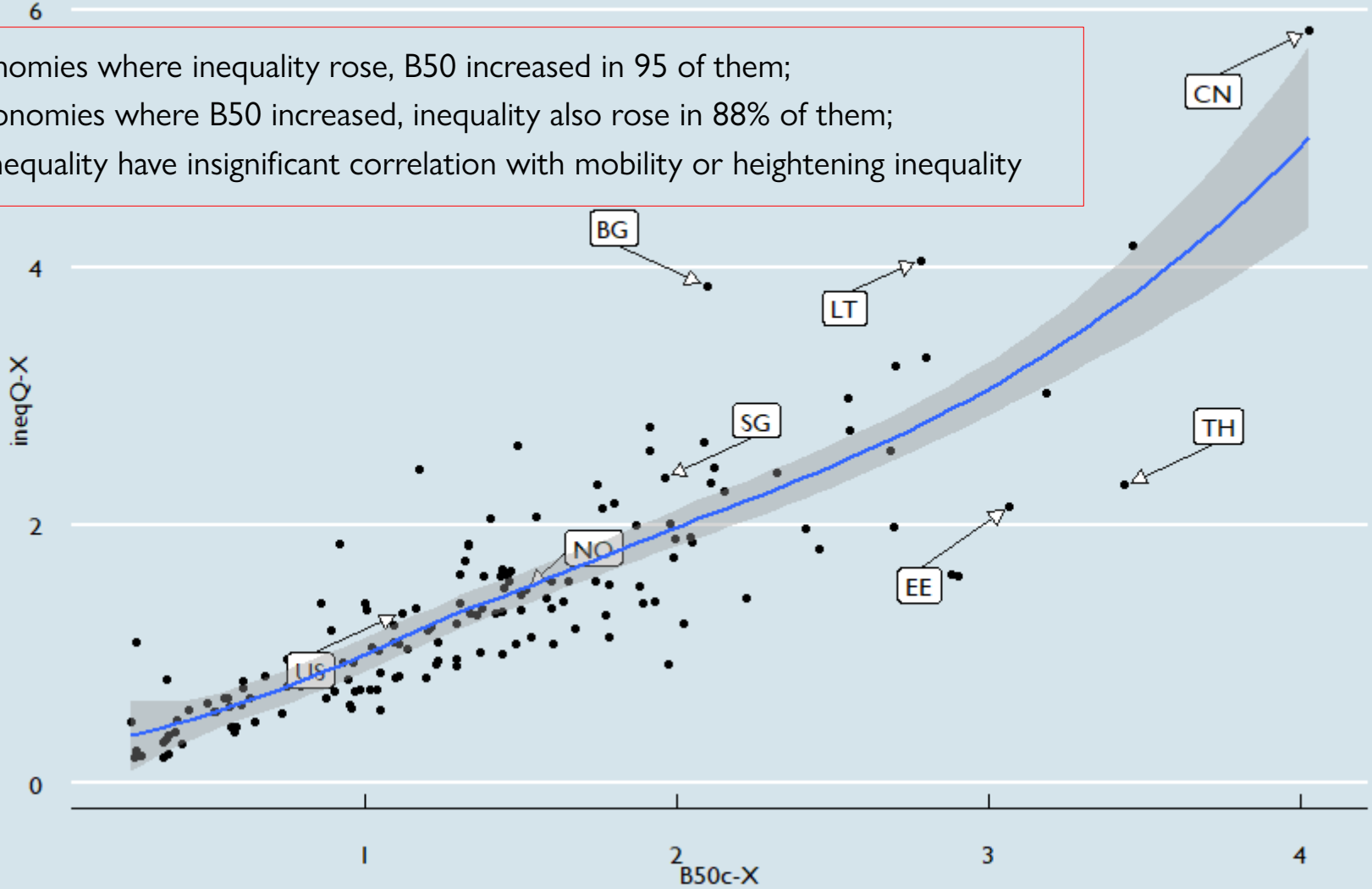
Correlation

		Correlation	95% Confidence Interval	t(154)
Income Mobility	**Inequality Rise**	0.84	(0.79, 0.88)	19.45
Income Mobility	**Inequality Level**	-0.03	(-0.19, 0.13)	-0.40
Inequality Rise	**Inequality Level**	0.01	(-0.15, 0.17)	0.14

1. Of 99 economies where inequality rose, B50 increased in 95 of them;
2. Of 108 economies where B50 increased, inequality also rose in 88% of them;
3. Levels of inequality have insignificant correlation with mobility or heightening inequality

2000-2019 When the bottom 50% rise, so too inequality

1. Of 99 economies where inequality rose, B50 increased in 95 of them;
2. Of 108 economies where B50 increased, inequality also rose in 88% of them;
3. Levels of inequality have insignificant correlation with mobility or heightening inequality



“Inequality Is No Sufficient Statistic”

Our greatest fears and concerns

Resource-Inadequacy vs Disparity

Empirical Evidence

Inequality Is No Sufficient Statistic

IEA RIDGE Lecture

by Danny Quah

Sat 03 Jul 2021

NUS National University of Singapore

Lee Kuan Yew School of Public Policy

Emerging Grand Narrative

"Globalisation and technological change have polarised society into a small elite with highly paid, secure jobs on one side, and on the other side are growing numbers of people, including an increasingly squeezed middle class, in insecure, poorly-paid work."

This growing inequality has resulted in "erosion of solidarity, social trust and faith in democratic institutions".

(Nolan, 2019)

What do people really understand about inequality and deprivation?

"New analysis ... showing that 14mn people live in poverty highlights just how unequal a society the UK has become."

"The poorer half of the population are as poor today as they were in the past, with barely 5 percent of total wealth in 2010, just as in 1910."

US Wealth Inequality 2005	Survey estimate share	Actual share
Top 20%	59%	84%
Bottom 40%	10%	0.3%

Source: Norton and Ariely 2011

Inequality? Or fairness? Or yet something else?

Political Upheaval

- Characteristics of Populism
  1. Ordinary people against elites;
  2. Opposition to current system (whatever that might be).
- Visibility:
  1. Decline in trust and perceived legitimacy of the system;
  2. In democracies, the rise of extremist parties

Political Upheaval – Previous Findings

1. Timeline for trust shows no clear trend in Europe (Gould and Hiczen 2016; Sarracino and Mikucka 2016).
2. Trust in the US explained not by overall inequality but by inequality among the bottom 50% (Gould and Hiczen 2016).
3. Voter shift toward extremism in democracies: More cultural attitude than economic variables (Inglehart and Norris 2016). Austria, France stable inequality; Poland strong income growth in the bottom 50%.
4. Mass protests and strikes worldwide: Aggregate recessions, growth slowdowns, worsening living conditions, deteriorating expectations of "a purposeful and meaningful life" (Witte et al 2020). Not inequality